Question 1

GH-300: Copilot Architecture

What fundamentally powers GitHub Copilot's code completion capabilities?

A. Static rule-based syntax analysis engines.

✔ **B. Large language models (LLMs) trained on billions of lines of code.**

C. Automated regex search index queries against public repositories.

D. Local machine learning models compiled directly into IDE binaries.

Correct Answer: B

GitHub Copilot is powered by generative Large Language Models (LLMs) fine-tuned on public code datasets. It analyzes editor context and uses probabilistic language modeling to suggest contextually relevant code syntax in real time.

Option Breakdown

- **A. Static rule-based syntax analysis engines:** Incorrect. Traditional linters use static rules; Copilot uses generative AI.
- **B. Large language models (LLMs):** Correct. Deep learning transformer models generate contextual completions.
- **C. Automated regex search index queries:** Incorrect. Copilot generates new code probabilistically rather than running regex lookup queries.
- **D. Local machine learning models:** Incorrect. Models run securely in the GitHub/Azure cloud infrastructure, not locally inside IDE binaries.

💡 **GH-300 Exam Tip**

Copilot processes requests via cloud services (Azure OpenAI Service). The IDE extension acts as a client that sends contextual prompts and receives generated completions back over HTTPS.